

```
Call: glm.nb(formula = count ~ year_centered * factor(case_sub_type) +
  factor(zip_code), data = df_counts, init.theta = 5.376004558,
  link = log)
```

Coefficients:

```
(Intercept)
0.0008780
year_centered
0.0283741
factor(case_sub_type)industrial_or_comm_equip
0.0282053
factor(case_sub_type)leaf_blower
-0.2991966
factor(case_sub_type)leaf_blower_gas
1.5644331
factor(case_sub_type)other_causes
-0.3191763
factor(case_sub_type)residential_construction
0.2009721
factor(case_sub_type)residential_mechanical
-0.7418655
factor(case_sub_type)residential_motor_vehicle
-0.8217784
factor(case_sub_type)residential_house_yard
-0.6872711
factor(case_sub_type)residential_multi_tenant_dwelling
-0.2525680
factor(zip_code)20083
0.2201402
factor(zip_code)20110
-0.0184771
factor(zip_code)20646
0.2269926
factor(zip_code)20705
-0.0722099
factor(zip_code)20707
-0.1157781
factor(zip_code)20743
-0.2151311
factor(zip_code)20770
-0.2017117
factor(zip_code)20774
-0.0591635
factor(zip_code)20782
-0.0369665
factor(zip_code)20812
0.0009783
factor(zip_code)20813
-0.0131944
```

```
factor(zip_code) 20814
1.9331605
factor(zip_code) 20815
1.3951571
factor(zip_code) 20816
1.0516681
factor(zip_code) 20817
1.1866593
factor(zip_code) 20818
-0.1192971
factor(zip_code) 20822
-0.4989447
factor(zip_code) 20827
-0.0264065
factor(zip_code) 20832
0.5474544
factor(zip_code) 20833
0.3885083
factor(zip_code) 20837
0.2513612
factor(zip_code) 20838
0.2418194
factor(zip_code) 20839
0.4473205
factor(zip_code) 20841
0.2256769
factor(zip_code) 20842
0.2203381
factor(zip_code) 20850
1.3431941
factor(zip_code) 20851
0.4029418
factor(zip_code) 20852
1.4363980
factor(zip_code) 20853
0.7568507
factor(zip_code) 20854
1.2006899
factor(zip_code) 20855
0.4364641
factor(zip_code) 20860
-0.1723598
factor(zip_code) 20861
0.0647115
factor(zip_code) 20862
-0.8854347
factor(zip_code) 20866
0.4116217
factor(zip_code) 20868
```

```
0.2132233
factor(zip_code) 20871
0.5606451
factor(zip_code) 20872
0.4260642
factor(zip_code) 20874
1.1924642
factor(zip_code) 20876
0.6932648
factor(zip_code) 20877
0.5530169
factor(zip_code) 20878
0.7821324
factor(zip_code) 20879
0.6961732
factor(zip_code) 20880
0.0981453
factor(zip_code) 20882
0.3869196
factor(zip_code) 20884
-0.0171564
factor(zip_code) 20885
-0.0131944
factor(zip_code) 20886
1.0642898
factor(zip_code) 20887
0.0771999
factor(zip_code) 20889
-0.0276341
factor(zip_code) 20891
-0.1091610
factor(zip_code) 20892
0.6876677
factor(zip_code) 20895
0.8618709
factor(zip_code) 20896
0.2124037
factor(zip_code) 20899
-0.0105530
factor(zip_code) 20901
0.9353450
factor(zip_code) 20902
1.1141465
factor(zip_code) 20903
0.3335263
factor(zip_code) 20904
0.9064682
factor(zip_code) 20905
0.4426983
```

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factor(zip_code)20906
1.0526565
factor(zip_code)20910
1.7997933
factor(zip_code)20912
0.3207717
factor(zip_code)20985
-1.1692796
factor(zip_code)21036
-0.1748730
factor(zip_code)21061
-0.2106580
factor(zip_code)21207
-0.1659267
factor(zip_code)21666
-0.0079117
factor(zip_code)21754
-0.1703999
factor(zip_code)21771
0.0763793
factor(zip_code)21816
-0.0131944
factor(zip_code)22101
-0.2106580
factor(zip_code)35512
0.1028111
factor(zip_code)77380
0.1713803
factor(zip_code)77386
0.1041086
factor(zip_code)88191
-0.0145150
factor(zip_code)92024
0.0283741
factor(zip_code)missing
1.8086877
year_centered:factor(case_sub_type)industrial_or_comm_equip
-0.0270534
year_centered:factor(case_sub_type)leaf_blower
-0.0662831
year_centered:factor(case_sub_type)leaf_blower_gas
NA
year_centered:factor(case_sub_type)other_causes
0.0168789
year_centered:factor(case_sub_type)residential_construction
-0.0239009
year_centered:factor(case_sub_type)residential_mechanical
-0.0403366
year_centered:factor(case_sub_type)residential_motor_vehicle

```

```
                                -0.0476504
      year_centered:factor(case_sub_type)residential_house_yard
                                -0.0699666
year_centered:factor(case_sub_type)residential_multi_tenant_dwelling
                                -0.0149197
```

```
Degrees of Freedom: 3780 Total (i.e. Null);  3684 Residual
Null Deviance:      5962
Residual Deviance: 2787  AIC: 14300
```